Simple LSTM Forward Pass with Full Intermediate Steps (5 Time Steps)

Gary Ewing

This version expands the previous example with every single arithmetic operation shown explicitly for each gate and state update. All calculations use the exact expanded forward-pass equations.

Weights (Randomly Initialized, Shared Across All Time Steps)

Using the naming convention from our previous work:

Forget gate

- $w_{hf} = 0.28$
- $w_{xf} = -0.15$
- $b_f = 0.05$

Input gate (sigmoid)

- $w_{hi1} = 0.35$
- $w_{xi1} = 0.42$
- $b_{i1} = 0.18$

Candidate

- $w_{hi2} = 0.22$
- $w_{xi2} = 0.31$
- $b_{i2} = 0.09$

Output gate

- $w_{ho} = 0.47$
- $w_{xo} = 0.19$
- $b_o = 0.26$

Input Sequence

- $x_1 = 0.5$
- $x_2 = 0.7$
- $x_3 = 0.4$
- $x_4 = 0.8$
- $x_5 = 0.6$

Initial State

- $h_0 = 0$
- $C_0 = 0$

The Six Equations Used at Every Time Step

$$\begin{aligned}
f_t &= \sigma(w_{hf} \cdot h_{t-1} + w_{xf} \cdot x_t + b_f) \\
i_t &= \sigma(w_{hi1} \cdot h_{t-1} + w_{xi1} \cdot x_t + b_{i1}) \\
\tilde{C}_t &= \tanh(w_{hi2} \cdot h_{t-1} + w_{xi2} \cdot x_t + b_{i2}) \\
o_t &= \sigma(w_{ho} \cdot h_{t-1} + w_{xo} \cdot x_t + b_o) \\
C_t &= f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \\
h_t &= o_t \odot \tanh(C_t)
\end{aligned}$$

(where $\sigma(z) = \frac{1}{1+e^{-z}}$ and tanh is the standard hyperbolic tangent function).

Full Forward Pass with Every Intermediate Step

Time Step $t = 1$

Inputs: $h_0 = 0$, $C_0 = 0$, $x_1 = 0.5$

Forget gate

- Linear combination: $0.28 \times 0 + (-0.15) \times 0.5 + 0.05 = -0.075 + 0.05 = -0.025$
- $f_1 = \sigma(-0.025) = \frac{1}{1+e^{0.025}} \approx \frac{1}{2.025315} \approx 0.4938$

Input gate (sigmoid)

- Linear combination: $0.35 \times 0 + 0.42 \times 0.5 + 0.18 = 0.21 + 0.18 = 0.39$
- $i_1 = \sigma(0.39) \approx 0.5963$

Candidate

- Linear combination: $0.22 \times 0 + 0.31 \times 0.5 + 0.09 = 0.155 + 0.09 = 0.245$
- $\tilde{C}_1 = \tanh(0.245) \approx 0.2404$

Output gate

- Linear combination: $0.47 \times 0 + 0.19 \times 0.5 + 0.26 = 0.095 + 0.26 = 0.355$
- $o_1 = \sigma(0.355) \approx 0.5878$

Cell state update

- $C_1 = (0.4938 \times 0) + (0.5963 \times 0.2404) = 0 + 0.1434 = 0.1434$

Hidden state update

- $h_1 = 0.5878 \times \tanh(0.1434) \approx 0.5878 \times 0.1426 = 0.0838$

Updated after $t = 1$: $h_1 \approx 0.0838$, $C_1 \approx 0.1434$

Time Step $t = 2$

Inputs: $h_1 \approx 0.0838$, $C_1 \approx 0.1434$, $x_2 = 0.7$

Forget gate

- Linear combination: $0.28 \times 0.0838 + (-0.15) \times 0.7 + 0.05 = -0.0315$
- $f_2 = \sigma(-0.0315) \approx 0.4921$

Input gate (sigmoid)

- Linear combination: $0.35 \times 0.0838 + 0.42 \times 0.7 + 0.18 = 0.5033$
- $i_2 = \sigma(0.5033) \approx 0.6232$

Candidate

- Linear combination: $0.22 \times 0.0838 + 0.31 \times 0.7 + 0.09 = 0.3254$
- $\tilde{C}_2 = \tanh(0.3254) \approx 0.3140$

Output gate

- Linear combination: $0.47 \times 0.0838 + 0.19 \times 0.7 + 0.26 = 0.4324$
- $o_2 = \sigma(0.4324) \approx 0.6064$

Cell state update

- $C_2 = (0.4921 \times 0.1434) + (0.6232 \times 0.3140) = 0.2663$

Hidden state update

- $h_2 = 0.6064 \times \tanh(0.2663) \approx 0.1578$

Updated after $t = 2$: $h_2 \approx 0.1578$, $C_2 \approx 0.2663$

Time Step $t = 3$

Inputs: $h_2 \approx 0.1578$, $C_2 \approx 0.2663$, $x_3 = 0.4$

Forget gate

- Linear combination: $0.28 \times 0.1578 + (-0.15) \times 0.4 + 0.05 = 0.0342$
- $f_3 = \sigma(0.0342) \approx 0.5086$

Input gate (sigmoid)

- Linear combination: $0.35 \times 0.1578 + 0.42 \times 0.4 + 0.18 = 0.4032$
- $i_3 = \sigma(0.4032) \approx 0.5995$

Candidate

- Linear combination: $0.22 \times 0.1578 + 0.31 \times 0.4 + 0.09 = 0.2487$
- $\tilde{C}_3 = \tanh(0.2487) \approx 0.2440$

Output gate

- Linear combination: $0.47 \times 0.1578 + 0.19 \times 0.4 + 0.26 = 0.4102$
- $o_3 = \sigma(0.4102) \approx 0.6011$

Cell state update

- $C_3 = (0.5086 \times 0.2663) + (0.5995 \times 0.2440) = 0.2817$

Hidden state update

- $h_3 = 0.6011 \times \tanh(0.2817) \approx 0.1649$

Updated after $t = 3$**:** $h_3 \approx 0.1649$, $C_3 \approx 0.2817$

Time Step $t = 4$

Inputs: $h_3 \approx 0.1649$, $C_3 \approx 0.2817$, $x_4 = 0.8$

Forget gate

- Linear combination: $0.28 \times 0.1649 + (-0.15) \times 0.8 + 0.05 = -0.0238$
- $f_4 = \sigma(-0.0238) \approx 0.4940$

Input gate (sigmoid)

- Linear combination: $0.35 \times 0.1649 + 0.42 \times 0.8 + 0.18 = 0.5737$
- $i_4 = \sigma(0.5737) \approx 0.6396$

Candidate

- Linear combination: $0.22 \times 0.1649 + 0.31 \times 0.8 + 0.09 = 0.3743$
- $\tilde{C}_4 = \tanh(0.3743) \approx 0.3580$

Output gate

- Linear combination: $0.47 \times 0.1649 + 0.19 \times 0.8 + 0.26 = 0.4895$
- $o_4 = \sigma(0.4895) \approx 0.6200$

Cell state update

- $C_4 = (0.4940 \times 0.2817) + (0.6396 \times 0.3580) = 0.3681$

Hidden state update

- $h_4 = 0.6200 \times \tanh(0.3681) \approx 0.2184$

Updated after $t = 4$: $h_4 \approx 0.2184$, $C_4 \approx 0.3681$

Time Step $t = 5$

Inputs: $h_4 \approx 0.2184$, $C_4 \approx 0.3681$, $x_5 = 0.6$

Forget gate

- Linear combination: $0.28 \times 0.2184 + (-0.15) \times 0.6 + 0.05 = 0.0212$
- $f_5 = \sigma(0.0212) \approx 0.5053$

Input gate (sigmoid)

- Linear combination: $0.35 \times 0.2184 + 0.42 \times 0.6 + 0.18 = 0.5084$
- $i_5 = \sigma(0.5084) \approx 0.6244$

Candidate

- Linear combination: $0.22 \times 0.2184 + 0.31 \times 0.6 + 0.09 = 0.3240$
- $\tilde{C}_5 = \tanh(0.3240) \approx 0.3130$

Output gate

- Linear combination: $0.47 \times 0.2184 + 0.19 \times 0.6 + 0.26 = 0.4766$
- $o_5 = \sigma(0.4766) \approx 0.6169$

Cell state update

- $C_5 = (0.5053 \times 0.3681) + (0.6244 \times 0.3130) = 0.3814$

Hidden state update

- $h_5 = 0.6169 \times \tanh(0.3814) \approx 0.2246$

Final updated values after $t = 5$: $h_5 \approx 0.2246$, $C_5 \approx 0.3814$

This completes the full forward pass. Every intermediate linear combination, activation, and state update has been shown with explicit arithmetic for all 5 time steps using the exact equations we derived. The states h_t and C_t are updated after each step and carried forward to the next time step.